

Vinothana Balakrishnan

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Summary

To secure an opportunity that leverages my strong foundation in IT and software development. I aim to gain practical experience and contribute to a dynamic team while enhancing my technical skills in C++, JavaScript, AI/ML and full-stack development. I am committed to continuous learning and preparing for a successful career in the IT industry.

Projects

Automated Fetal Growth Monitoring using Ultrasound Images

🐙 Fetal Monitoring

- Built a **CNN-based regression model** to predict fetal head circumference from ultrasound images and estimate gestational age for fetal growth monitoring.

Tech Stack: Deep Learning, CNN Regression, Medical Image Processing, PyTorch, OpenCV, Matplotlib

Multilingual Fake News detection

🐙 RealYaRubbish

- Developed a multilingual NLP-based fake news classification system supporting English, Hindi, and Tamil. Implemented TF-IDF feature extraction with multiple ML classifiers and ensemble voting for robust real-time prediction.

Tech Stack: Python, scikit-learn, Pandas, NumPy, NLTK, langdetect, TF-IDF Vectorizer, Streamlit.

TNEA COLLEGE SEARCH

🐙 Know Your College

- Built a full-stack web application to assist engineering aspirants with college prediction using cutoff and rank-based filtering. Implemented secure authentication, role-based access, RESTful APIs, optimized MySQL queries, and XLSX data export.

Tech Stack: React.js, Node.js, Express.js, MySQL, Bcrypt, ExcelJS.

SQL VISUALIZER

🌐 Website 🐙 SEEQL

- Designed an interactive SQL learning platform with animated visualizations to demonstrate query execution, joins, and subqueries. Implemented state-driven animations and modular React components to simplify database concepts.

Tech Stack: React.js, Javascript, HTML, CSS

Skills

Programming Languages: C++, C, Python

Full Stack: HTML, CSS, Javascript, ReactJS, NodeJS, ExpressJS

Frameworks/libraries: Scikit-learn, Mediapipe, Streamlit, OpenCV, Axios, ExcelJS, NLTK, Langdetect

Database: MySQL

Tools: Git, VS Code, Jupyter, Postman

Education

Velammal Institute of Technology, BE in Computer Science and Engineering

Sept 2023 – Present

- CGPA: 9.04/10

Experiences and Participations

Open Source Contributor – GSSoC'25

- Selected as a contributor for GSSoC 2025. Worked with technologies like JavaScript, ReactJS.

Tech Contributor – Hactoberfest'25

- Contributed production-level code to open-source repositories. Improved project performance and implemented new features

Hackathon Participation

- Participated in Smart India Hackathon 2024 and 25- Selected in the internal round